

SIOUX FALLS 14 NNE, SD, USA

WMO: 726600

Lat: 43.7346N

Lon: 96.6222W

Elev: 1594

StdP: 13.87

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 04990

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-7.1	-17.3	2.3	-11.4	-12.9	2.9	-6.9	19.5	17.8	17.5	19.6	4.6	N/A	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	89.7	73.5	86.7	72.2	84.0	70.4	77.4	85.5	75.3	83.9	73.4	81.4	7.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.1	139.6	82.8	72.8	129.1	80.6	70.7	120.1	78.8	42.1	85.3	39.9	84.2	38.1	81.7	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.4	14.3	12.5	DB	-20.0	95.1	5.8	2.8	-24.2	97.2	-27.6	98.8	-30.9	100.4	-35.1	102.5	(4)
			WB	-20.4	80.2	5.9	1.9	-24.6	81.6	-28.1	82.7	-31.4	83.8	-35.6	85.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	15.8	18.1	33.1	45.9	57.7	67.8	72.1	69.1	62.2	47.8	33.7	20.3
	DBStd	22.22	13.01	12.33	13.03	10.55	8.88	6.27	5.53	5.34	7.90	9.72	10.67	11.61
	HDD50	4297	1059	892	539	195	31	0	0	0	7	160	494	920
	HDD65	7737	1524	1312	989	575	259	42	9	22	143	539	938	1385
	CDD50	2642	0	0	16	73	269	535	686	592	374	91	6	0
	CDD65	608	0	0	1	3	33	127	230	149	60	5	0	0
	CDH74	5608	0	0	9	76	415	1168	2128	1170	569	73	0	0
	CDH80	1707	0	0	1	22	125	345	733	315	152	14	0	0
Wind	WSAvg	5.4	6.0	6.1	6.4	6.8	6.5	5.1	4.1	3.9	4.6	5.3	5.3	5.3
Precipitation	PrecAvg	27.4	0.6	0.8	1.5	2.9	3.7	4.4	3.3	3.3	2.7	2.2	1.2	0.8
	PrecMax	37.1	1.4	2.2	4.1	6.8	8.0	12.9	8.3	7.8	7.3	6.1	4.3	2.1
	PrecMin	17.5	0.0	0.2	0.2	0.5	1.0	1.0	0.4	1.1	0.2	0.3	0.0	0.1
	PrecStd	5.0	0.3	0.5	1.0	1.4	1.9	2.2	2.2	1.9	1.8	1.7	1.2	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.5	55.1	73.2	83.6	89.5	92.2	95.4	91.0	88.7	82.0	67.6	50.2
		MCWB	39.5	48.0	54.6	61.7	64.8	73.3	75.6	75.5	71.7	61.1	54.4	43.3
	2%	DB	39.8	44.2	64.3	74.5	83.3	87.2	90.4	86.7	84.5	74.5	60.0	43.5
		MCWB	35.2	38.7	53.0	55.7	64.2	69.6	76.1	74.5	69.7	58.3	49.0	37.9
	5%	DB	36.1	38.9	57.4	68.8	78.5	84.0	87.1	83.5	80.7	68.8	53.8	38.2
		MCWB	33.5	35.3	50.1	53.3	62.3	69.0	74.4	72.1	66.8	55.5	45.4	34.8
	10%	DB	33.2	34.8	51.2	63.5	73.6	80.6	83.9	80.3	76.7	63.4	49.0	35.0
		MCWB	31.5	32.4	45.5	50.3	60.1	67.3	72.1	70.2	65.5	52.5	43.1	32.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.1	49.6	60.7	61.5	70.0	77.9	81.0	78.9	74.8	64.3	54.3	46.1
		MCDB	47.0	56.6	69.1	74.7	79.4	88.2	89.3	87.2	83.9	75.7	63.6	49.8
	2%	WB	35.8	40.3	55.2	58.2	67.1	74.7	78.3	76.3	72.4	60.1	49.9	38.4
		MCDB	38.6	44.8	62.5	69.7	79.8	84.7	86.4	84.1	80.8	69.6	57.4	42.9
	5%	WB	34.1	36.2	51.0	54.5	64.8	72.4	76.3	74.2	70.5	57.4	46.1	35.6
		MCDB	36.5	39.5	59.1	63.7	75.9	81.4	84.6	81.3	77.4	65.6	52.9	38.9
	10%	WB	32.1	32.9	45.8	51.4	62.4	69.9	74.3	72.0	68.1	54.3	42.4	33.2
		MCDB	33.9	35.2	52.8	59.8	71.3	78.9	82.7	78.7	74.1	62.1	48.1	35.3
Mean Daily Temperature Range	5% DB	MDBR	16.4	17.2	18.2	21.2	21.0	19.9	20.8	20.0	22.0	20.4	19.1	15.7
		MCDBR	19.9	22.0	28.7	31.7	28.1	24.0	23.7	23.0	26.1	28.3	26.7	20.9
		MCWBR	16.4	16.5	18.0	16.9	13.0	11.7	10.8	11.0	12.3	15.1	16.6	15.7
	5% WB	MCDBR	18.4	21.5	28.8	27.0	23.5	21.7	21.0	20.7	21.8	24.9	24.3	20.2
		MCWBR	15.2	16.2	17.8	16.1	11.8	12.2	11.3	11.4	12.2	14.4	15.6	15.8
Clear-Sky Solar Irradiance	taub	0.269	0.282	0.316	0.368	0.384	0.394	0.400	0.403	0.355	0.324	0.291	0.272	
	taud	2.432	2.416	2.409	2.291	2.324	2.352	2.333	2.299	2.413	2.473	2.482	2.439	
	Ebn at Noon	277	292	293	284	282	278	275	270	276	272	266	264	
	Edh at Noon	23	28	32	40	40	39	39	39	32	26	22	21	
All-Sky Solar Radiation	RadAvg	601	936	1222	1540	1774	1987	2113	1745	1388	937	645	478	
	RadStd	55	74	100	141	130	129	78	134	103	112	46	40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page